

Vectors and Matrices

Vectors

A **coordinate system** is defined by an origin point and the orientation and scale of a set of coordinate axes.

A **point** $p = (x, y)$ refers to a location in space, specified relative to a coordinate system.

A **vector** $v = \langle m, n \rangle$ refers to a **displacement**—an amount of change in each coordinate—and is typically drawn as an arrow pointing along the direction of displacement. The figure below shows three (3) vectors:

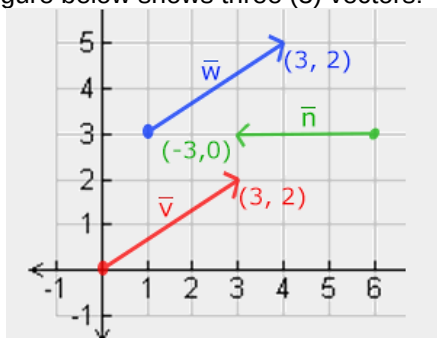


Figure 1. A collection of vectors.

- **Initial point** or **tail** – the point where the arrow begins
- **Terminal point** or **head** – the point where the arrow ends; indicates the result when the displacement has been applied to the initial point.
- **Length** or **magnitude** – the distance between the initial and terminal points of the vector; can be calculated from the components of the vector.

Vectors are not associated with any particular location in space; the same vector may exist at different locations. A vector whose initial point is located at the origin (when a coordinate system is specified) is said to be in **standard position**.

Individual numbers (that are not part of a point or vector) are often called **scalars**.

A variety of mathematical operations can be performed with and upon vectors. The first operation is the addition of vectors. Given vectors $v = \langle v_1, v_2 \rangle$ and $w = \langle w_1, w_2 \rangle$, the formula shall be:

$$v + w = \langle v_1 + w_1, v_2 + w_2 \rangle$$

Sample Vector Addition:

$$v = \langle 2, 5 \rangle$$

$$w = \langle 4, -2 \rangle$$

$$v + w = \langle v_1 + w_1, v_2 + w_2 \rangle$$

$$v + w = \langle 2 + 4, 5 + (-2) \rangle$$

$$v + w = \langle 6, 3 \rangle$$

The result is a new vector $u = v + w$ that shares the initial point of v and the terminal point of w .

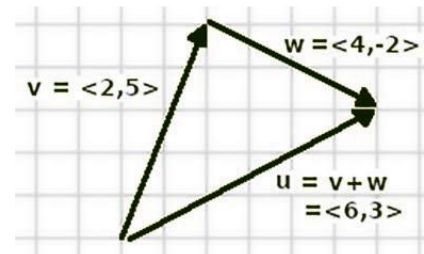


Figure 2. Vector addition.

Vector subtraction is the same as addition with a negated second vector:

$$v = \langle 1, 2 \rangle$$

$$w = \langle 4, 5 \rangle$$

$$v + -w = \langle v_1 + (-w_1), v_2 + (-w_2) \rangle$$

$$v + -w = \langle 1 + (-4), 2 + (-5) \rangle$$

$$v + -w = \langle -3, -3 \rangle$$

Given the length of the two (2) sides (x , y), you can determine the length/magnitude of a vector using the **Pythagorean theorem**.

$$||v|| = \sqrt{x^2 + y^2}$$

For example, the following computes the length of $v = \langle 4, 2 \rangle$:

$$||v|| = \sqrt{4^2 + 2^2} = \sqrt{16 + 4} = \sqrt{20} = 4.47$$

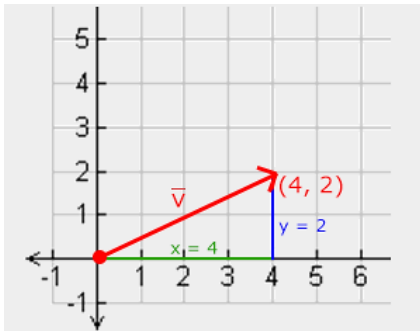


Figure 3. Sample graph showing a vector's length.

Matrices

A **matrix** is a rectangular array of values called **elements** that are typically accessed by means of **subscripts**. The first subscript refers to the row number, while the second one refers to the column number, with the subscripts starting at 0. The figure below shows an example of a 4x4 matrix.

$$\begin{bmatrix} A_{00} & A_{01} & A_{02} & A_{03} \\ A_{10} & A_{11} & A_{12} & A_{13} \\ A_{20} & A_{21} & A_{22} & A_{23} \\ A_{30} & A_{31} & A_{32} & A_{33} \end{bmatrix}$$

Figure 4. A 4x4 matrix.

The **identity matrix** contains all zeros, with ones along the diagonal. Any point or matrix multiplied by the identity matrix is unchanged.

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Figure 5. A 4x4 identity matrix.

The **transpose** of a matrix is computed by interchanging its rows and columns.

$$\begin{bmatrix} A_{00} & A_{01} & A_{02} & A_{03} \\ A_{10} & A_{11} & A_{12} & A_{13} \\ A_{20} & A_{21} & A_{22} & A_{23} \\ A_{30} & A_{31} & A_{32} & A_{33} \end{bmatrix} = \begin{bmatrix} A_{00} & A_{10} & A_{20} & A_{30} \\ A_{01} & A_{11} & A_{21} & A_{31} \\ A_{02} & A_{12} & A_{22} & A_{32} \\ A_{03} & A_{13} & A_{23} & A_{33} \end{bmatrix}$$

Figure 6. A transpose of a 4x4 matrix.

Matrices can be added or subtracted only when they have the same number of dimensions. You can find the sum simply by adding the corresponding entries in matrices.

$$\begin{aligned} A + B &= \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} \\ &= \begin{bmatrix} 1+5 & 2+6 \\ 3+7 & 4+8 \end{bmatrix} \\ &= \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix} \end{aligned}$$

The same rule applies to matrix subtraction:

$$\begin{aligned} A - B &= \begin{bmatrix} 4 & 2 \\ 1 & 6 \end{bmatrix} - \begin{bmatrix} 2 & 4 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 4-2 & 2-4 \\ 1-0 & 6-1 \end{bmatrix} \\ &= \begin{bmatrix} 2 & -2 \\ 1 & 5 \end{bmatrix} \end{aligned}$$

You can multiply a scalar by each element of the matrix using the dot operator. This is known as **scalar multiplication**.

$$2 \cdot \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 2 \cdot 1 & 2 \cdot 2 \\ 2 \cdot 3 & 2 \cdot 4 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}$$

In **matrix multiplication**, you can only multiply matrices if the number of columns on the left-hand side matrix is equal to the number of rows on the right-hand side matrix. To multiply matrices, multiply the elements of each row of the first matrix by the elements of each column in the second matrix. Then, add the products.

$$\begin{aligned} A \cdot B &= \begin{bmatrix} 3 & 2 & 1 & 5 \\ 9 & 1 & 3 & 0 \end{bmatrix} \cdot \begin{bmatrix} 2 & 9 & 0 \\ 1 & 3 & 5 \\ 2 & 4 & 7 \\ 8 & 1 & 5 \end{bmatrix} \\ A \cdot B &= \begin{bmatrix} 6+2+2+40 & 27+6+4+5 & 0+10+7+25 \\ 18+1+6+0 & 81+3+12+0 & 0+5+21+0 \end{bmatrix} \\ &= \begin{bmatrix} 50 & 42 & 42 \\ 25 & 96 & 26 \end{bmatrix} \end{aligned}$$

References:

Korites, B. (2018). *Python graphics: A reference for creating 2D and 3D images*. Apress.
 Marschner, S. & Shirley, P. (2021). *Fundamentals of computer graphics* (5th ed.). CRC Press.
 Stemkoski, L. & Pascale, M. (2021). *Developing graphics frameworks with Python and OpenGL*. CRC Press.